

HAT-P-36b

R-0762

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-36_260203 · created 2026-07-28T04:11:35+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2461074.9981 +/- 0.0052 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.107 +/- 0.023
Transit depth (Rp/Rs)^2	1.15 +/- 0.49 [%]
Orbital Inclination (inc)	84.51 +/- 2.51
Airmass coefficient 1 (a1)	0.991 +/- 0.013
Airmass coefficient 2 (a2)	0.007 +/- 0.061
Scatter in the residuals of the lightcurve fit is	1.29 %
Best Comparison Star	#1 - [98, 42]
Optimal Method	PSF photometry
Transit Duration (day)	0.0874 +/- 0.008

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.107 ± 0.023 vs 0.126 (deviation 0.83σ)
Duration fit vs published	0.0874 d vs 0.0925 d (deviation 0.64σ)
Mid-time O-C	-0.7 min
Depth SNR	4.7
Residual scatter	1.29 %

Light curve

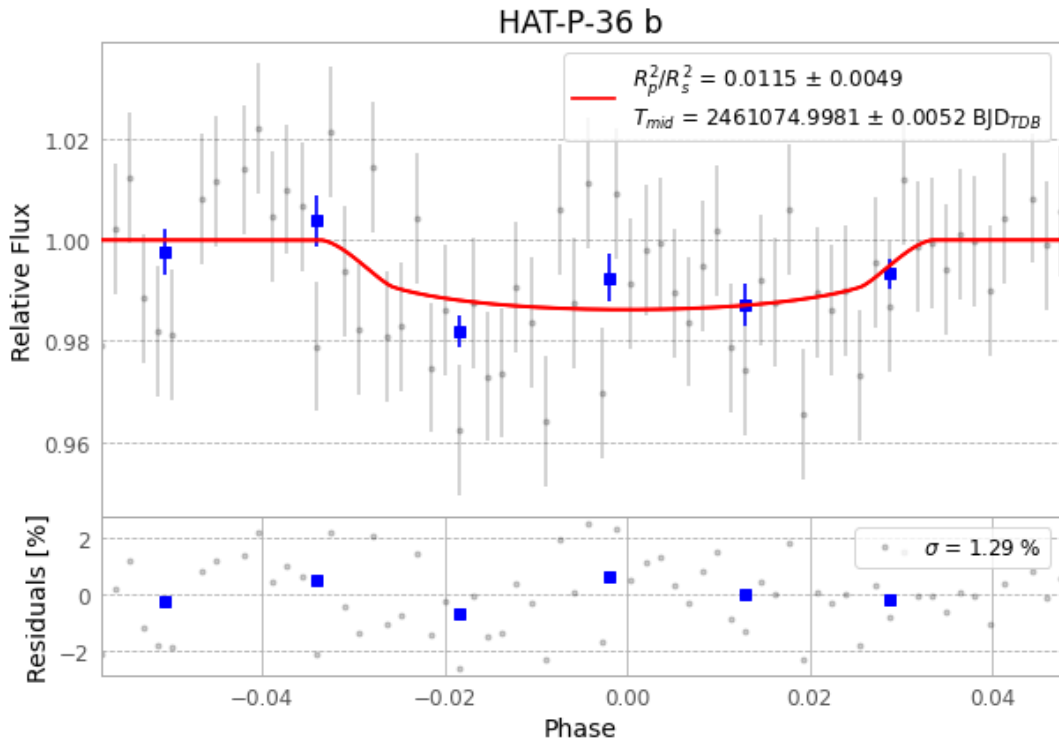


Figure 1 — FinalLightCurve_HAT-P-36 b_2026-02-03.png (SHA-256 f48646d7086e9c5a...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [370, 107], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 19135d4d529eaceb...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

68 FITS frames (45 MB), HATP-36260203100015.FITS → HATP-36260203132115.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-36-260203.log (4e526d8be4de...), FinalLightCurve_HAT-P-36 b_2026-02-03.png (f48646d7086e...), FinalLightCurve_HAT-P-36 b_2026-02-03.csv (5f71258dd797...)

Compute resources

Wall time 100.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-28 04:11Z.